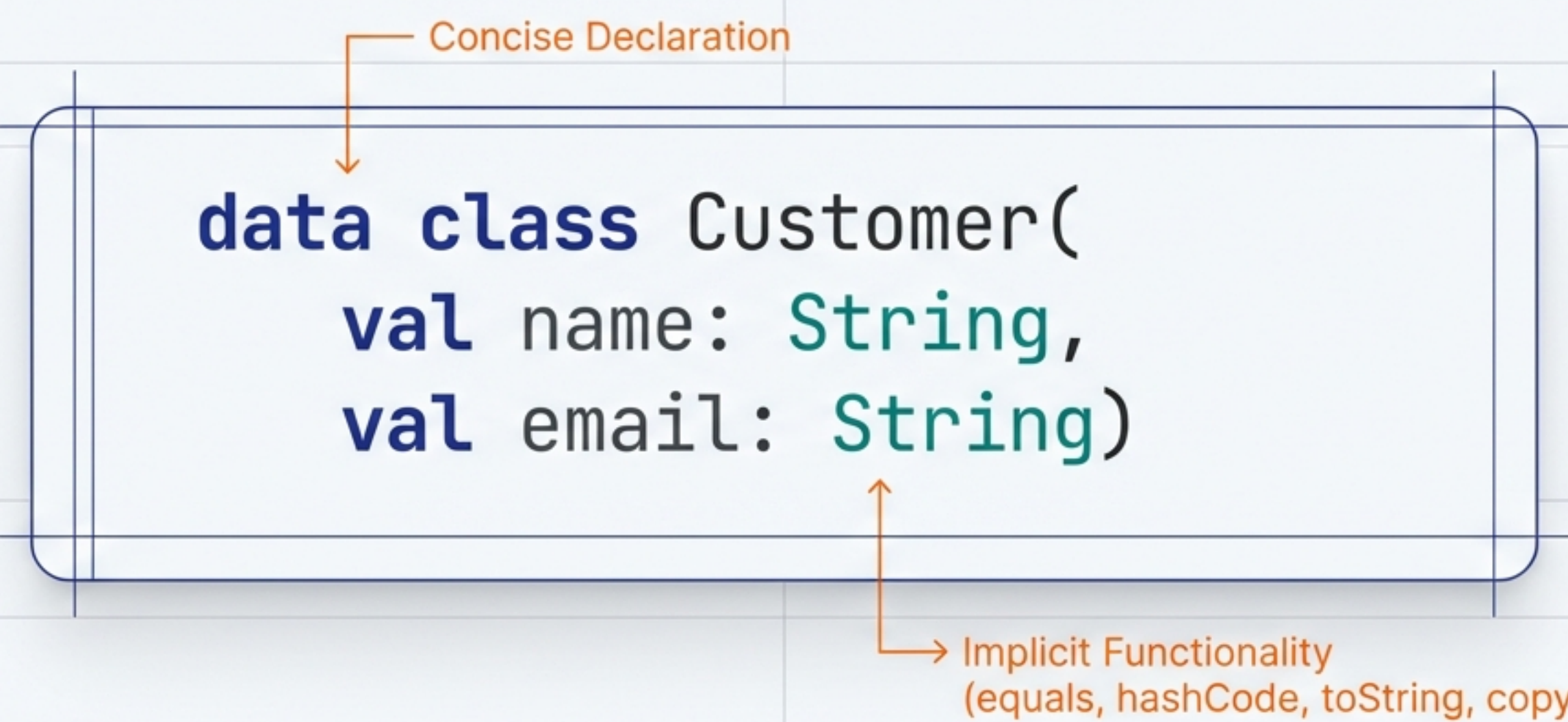


Idiomatic Kotlin: The Art of Concise Code

Essential patterns for clean, expressive, and safe development.



```
data class Customer(  
    val name: String,  
    val email: String)
```

Implicit Functionality
(equals, hashCode, toString, copy)

ADVANCED IDIOMS FOR 6TH-SEMESTER CS STUDENTS

From Syntactic Sugar to Semantic Power

Idioms in Kotlin aren't just shortcuts; they reduce cognitive load by removing the 'noise' of implementation details.

**The How
(Ceremony)**
in JetBrains Mono

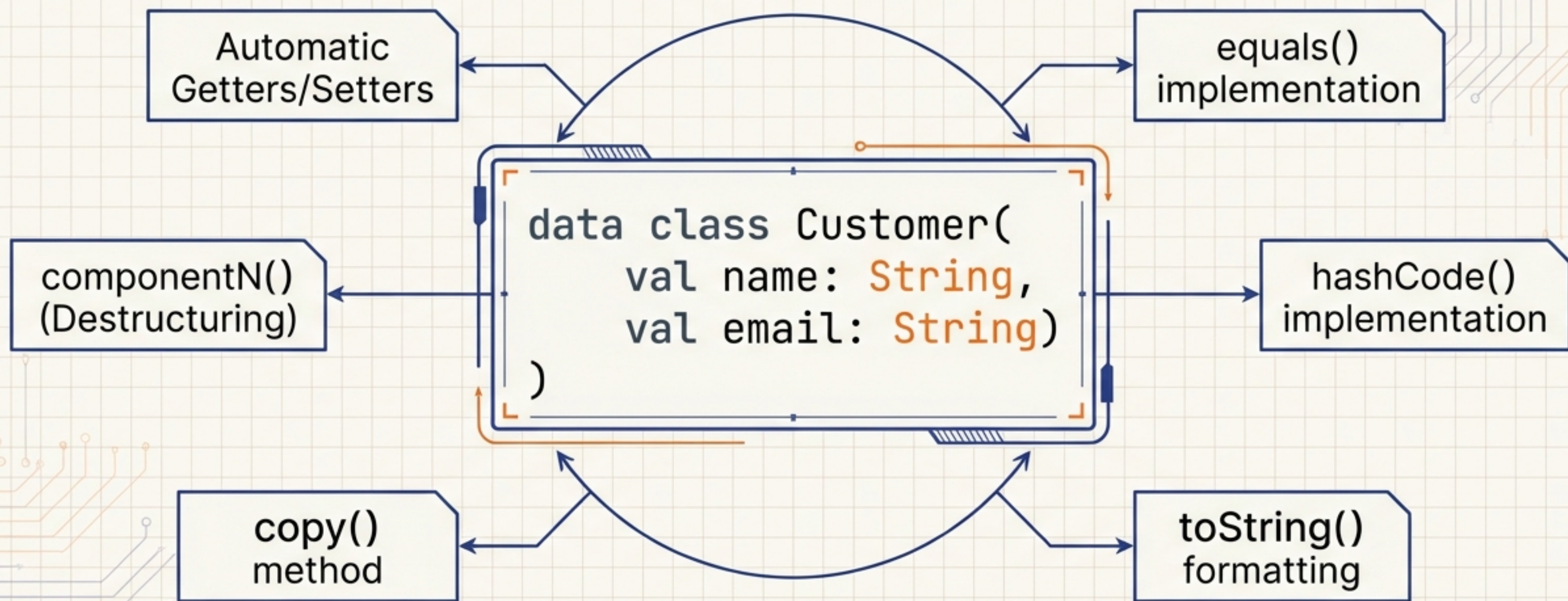
```
List<User> activeUsers = new ArrayList<>();  
for (User user : users) {  
    if (user.isActive()) {  
        activeUsers.add(user);  
    }  
}
```

val activeUsers = users.
filter { **it.isActive**
}

**The What
(Intent)**
in JetBrains Mono

Key Takeaway: Write code that reads like natural language. Focus on **intent**, not **ceremony**.

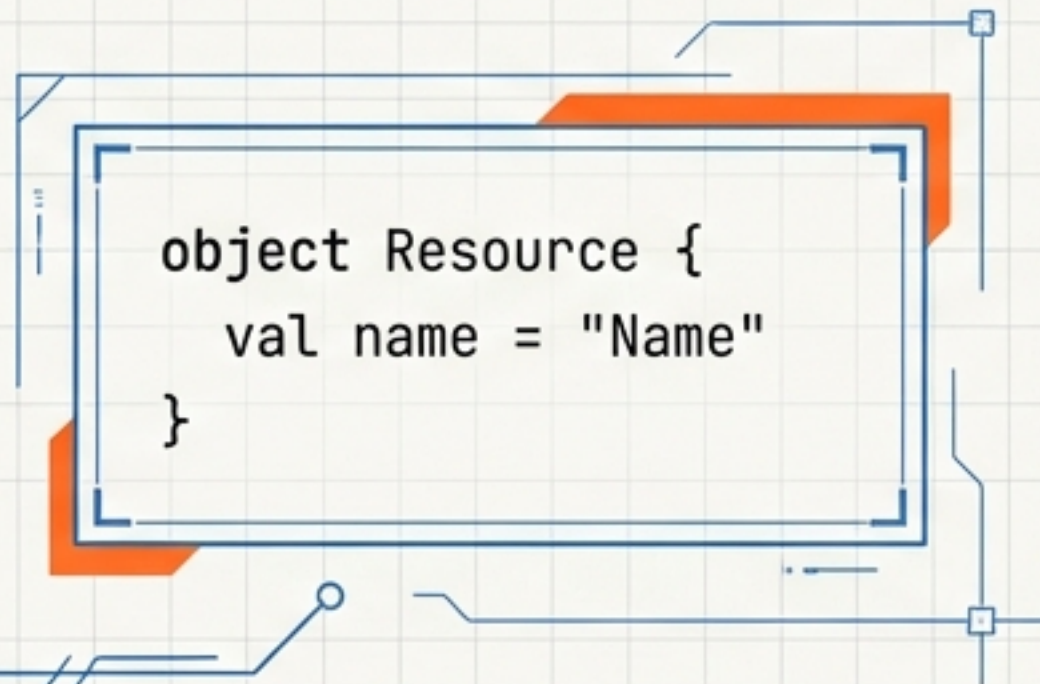
The Data Class Revolution



Replaces the entire DTO/POJO/Java Bean specification in a single line.

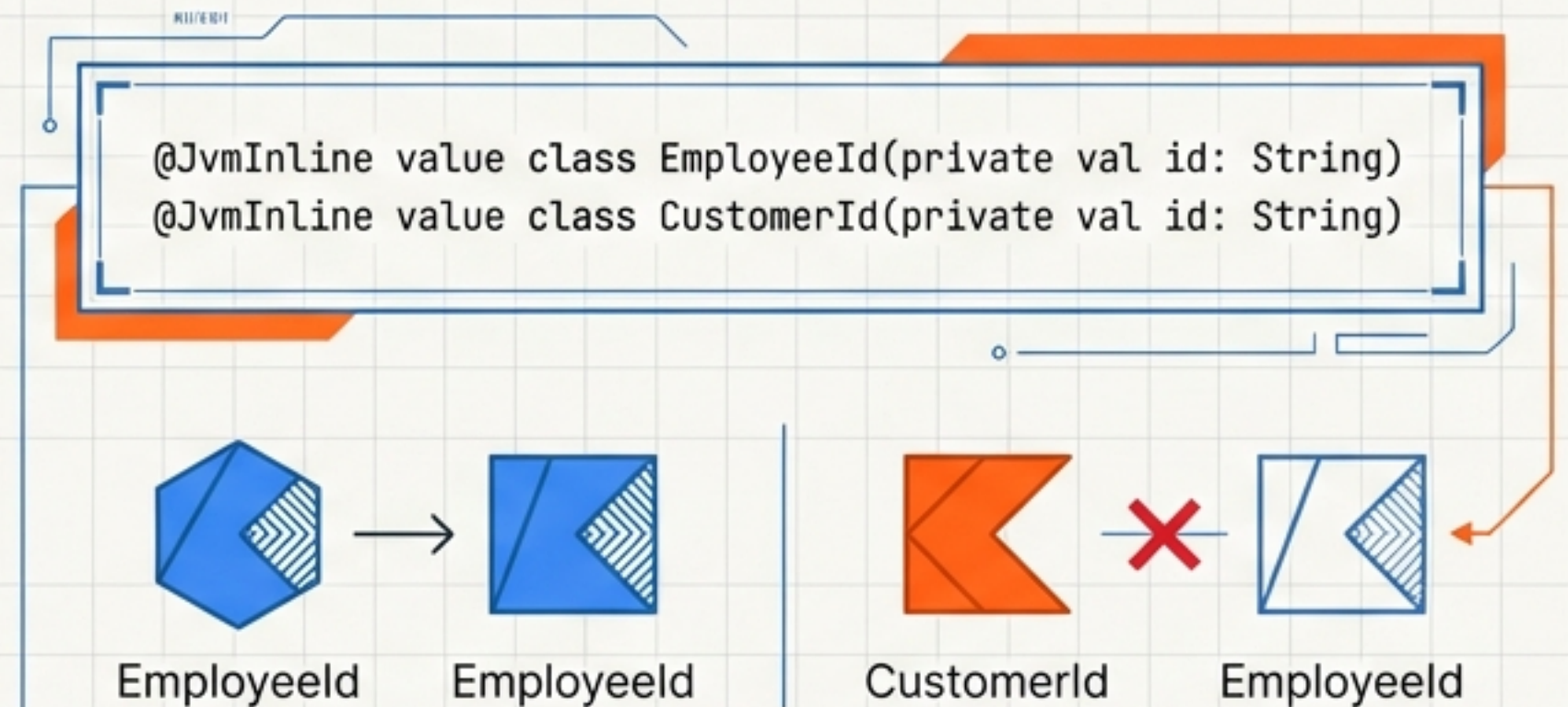
Objects and Zero-Overhead Safety

Thread-Safe Singletons



Lazy initialization and thread safety handled automatically.

Inline Value Classes



Compilation error if mixed up.
Zero runtime overhead.

Default Arguments & Interpolation

```
fun foo(a: Int = 0, b: String = "") {  
    ...  
}
```

Eliminates Method Overloading

```
println("Name: $name, Age: ${user.age}")
```

Direct Interpolation. No StringBuilder.

Expressions Over Statements

In Kotlin, control structures return values. Assign logic directly.

val status



```
if (users.isEmpty()) {  
    "No Users"  
} else {  
    "Active"  
}
```

fun isAlive()



health > 0

val result



try { count() } catch (e: Exception) { -1 }

The Power of when

```
val distinctValue = when (input) {  
  is String -> input.length // Smart Cast  
  "Red" -> 0  
  in 1..10 -> 100  
  else -> -1  
}
```

Automatic
Smart Cast

Pattern
Matching

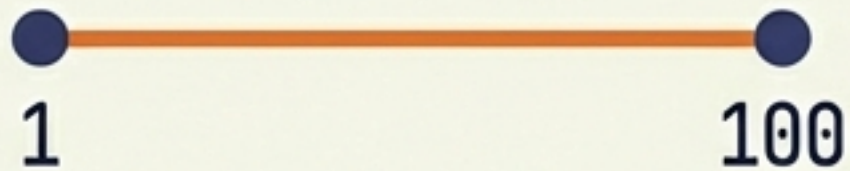
Exhaustiveness
Check

Returns a value
(Expression)

Looping & Ranges

Ranges

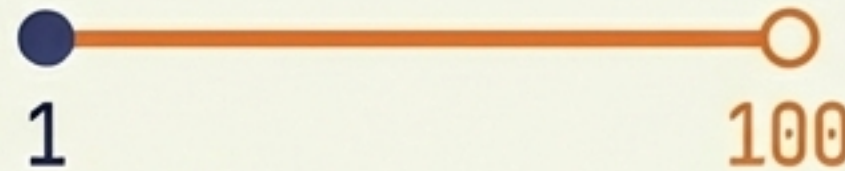
```
for (i in 1..100)
```



Closed Range

Until

```
for (i in 1..<100)
```



Open Ended

Maps

```
for ((k, v) in map)
```

↖
Destructuring
Declaration

Collections are immutable by default: `listOf()`, `mapOf()`.

Null Safety: The Billion Dollar Fix

→ The Problem
File might be missing?



`files?.size`

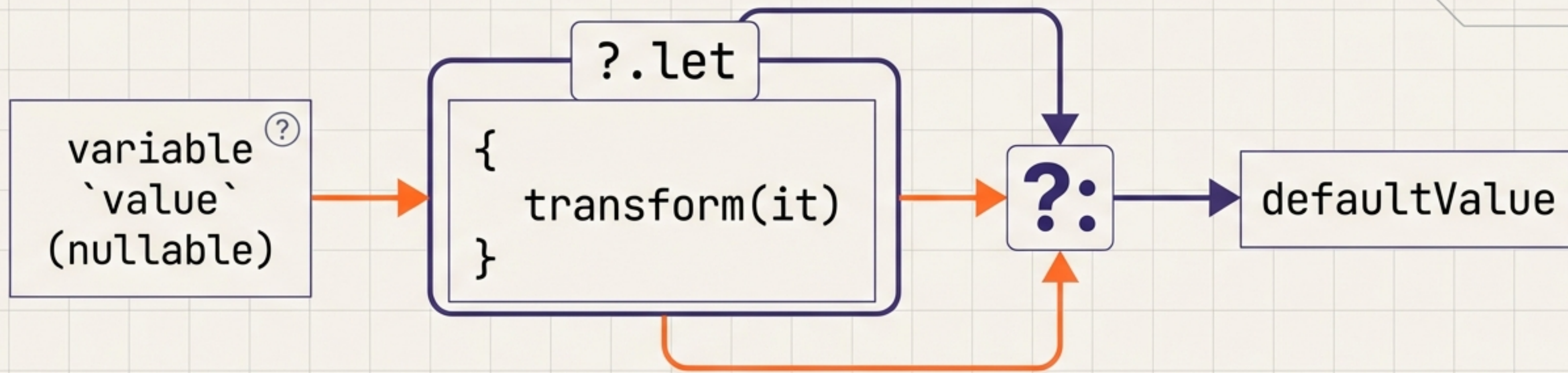
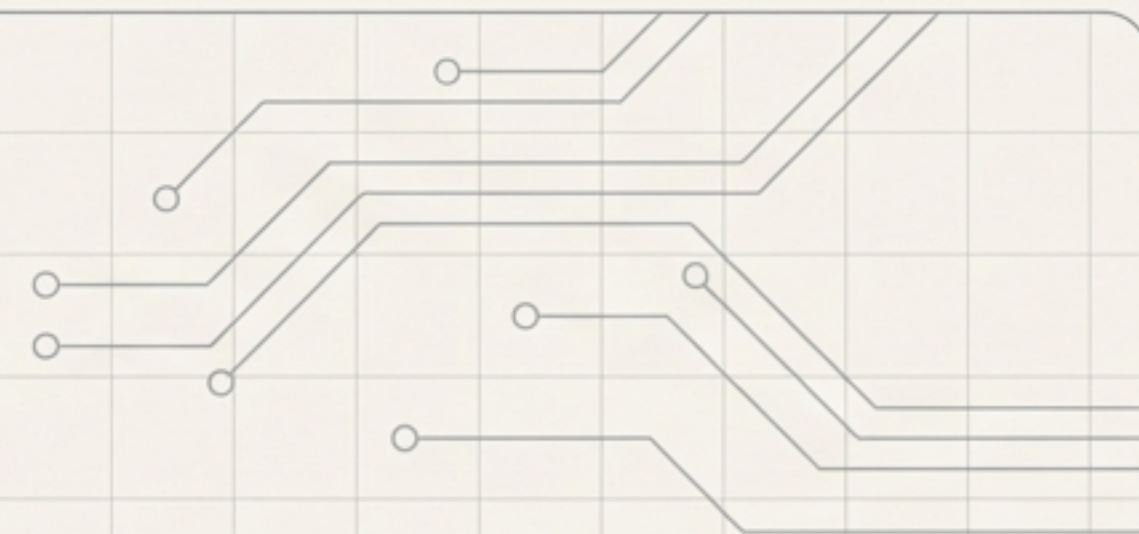
Safe Call (?). Returns null if files is null. No Crash.

`files?.size ?: 0`



Elvis Operator (?:). If result is null, use 0.

Safe Execution Idioms



```
val result = value?.let { transform(it) } ?: defaultValue
```

Replaces nested `if (x != null)` checks with fluent chains.

Functional Collection Operations

```
val positives = list.filter { it > 0 }
```

Higher-order
function.

Implicit name for single parameter.
No need for ``x -> x > 0``.

```
val hasEmail = "john@example.com" in emails
```

``in`` operator delegates
to ``contains()``.

Context Configuration (Scope Functions)

Initialize & Return

```
1 val rect = Rectangle().apply {  
2     length = 4  
3     color = 0xFAFAFA  
4 }
```

this = Rectangle

Group Operations

```
1 with(turtle) {  
2     penDown()  
3     forward(100.0)  
4 }
```

Context: turtle

Extensions & Lazy Evaluation

Lazy Evaluation

```
val heavyConfig by lazy {  
    loadConfig()  
}
```



Computed only
on first access.
Thread-safe.

Extensions

```
fun String.toCamelCase() { ... }  
"my text".toCamelCase()
```

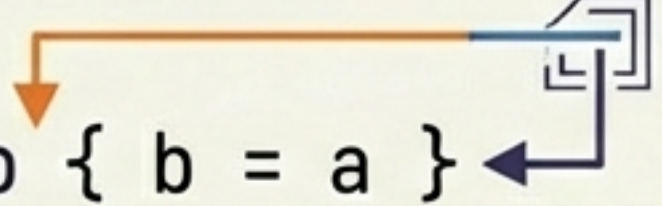


Extend classes you don't
own without inheritance.

The Developer's Toolkit

One-line Swap

```
a = b.also { b = a }
```

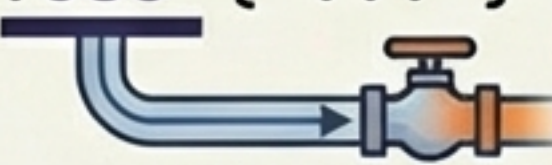


Standard Placeholder (Throws Error)

```
fun logic() =  
  TODO("Wait for accounting")
```

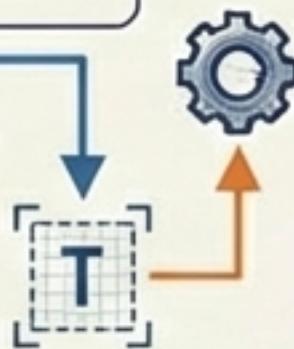
Auto-close Resources (Try-with-resources)

```
stream.reader().use { ... }
```



Access Generic Type T at Runtime

```
inline fun <reified T>  
  parse(json: String)
```



Practice the Idioms

```
> Refactor.kt
```

Idiomatic Kotlin is about clarity, not just brevity.
You have the tools; now solve the problems.

Challenge: Solve Advent of Code puzzles using
apply, filter, and when.